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AI-Driven Internship Matching Using Skill-Based Recommendation System

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Abstract* — The most common way of matching students/interns and job seekers with internships is a manual and time consuming process and not very efficient. Most of the time, the candidates do not match the skills required for the jobs posted and hence there is a huge mismatch. In this paper, we propose an internship matching system using Artificial Intelligence (AI) that makes use of skill based recommendation algorithms to bridge the gap. Our proposed system uses Natural Language Processing (NLP) techniques to extract and normalize the required skills in the resumes of the candidates and the job descriptions of the internships. It then uses a combination of content based filtering, collaborative filtering and hybrid recommendation approaches to compute a multi dimensional skill similarity score. These experimental models are trained on the historical data of the placements and are designed to maximize the match accuracy, and in addition, to be dynamic and adaptive to the changing skill requirements of the industry. This research work also contributes to the use of AI in career services and recruitment, which will lead to a fairer, more efficient and data-driven internship recruitment process.

Keywords*—Artificial Intelligence, Internship Recommendation, Skill Matching, Natural Language Processing, BERT, Hybrid Recommendation System, Machine Learning.

I. INTRODUCTION

From academic to the industry is probably one of the biggest steps in one's career. In order to get there, a common approach that students have, is doing internships. The internships are then a stepping stone for the practical knowledge, the professional networking and for converting the internship into full time job [1]. However, as the number of applicants as well as job postings have increased many folds, the existing keyword based approach to resume screening is proven to be not working effectively and efficiently to match the candidate with the required skills to the job posting.

All the modern career platforms like LinkedIn, Handshake, Internshala etc., follow the keyword matching and the shallowness of profile to description matching approach. These approaches do not match the semantic depth of the competency profiles of the candidates. For example, a candidate having ReactJS as a skill will not be matched in the search results for "Frontend Framework Developer" job even though he has the exact required competency – a phenomenon known as the "lexical gap" [2].

Current state of the art practices involving the matching of career learning resources with learners' skills, knowledge and preferences do not yet fully take advantage of the benefits brought by Artificial Intelligence (AI), Machine Learning (ML) or transformer-based advances in Natural Language Processing (NLP). Recent experiments carried out on the SkillRec [3] approach however show that by performing a semantic match at the skill level, the resulting accuracy is much higher. This paper therefore reviews the latest scientific works from 2022 to 2025 on the matching realised by the recent AI-based systems proposed for this task, and presents a hybrid matching approach that is the object of a proposed conceptual model for resources matching. The latter is based on extracting skills and preferences of learners using (NLP) techniques and combining them with content-based filtering (CBF), collaborative filtering (CF) based on knowledge graph reasoning and a reinforcement learning feedback loop.

II. LITERATURE REVIEW

A. SkillRec and Skill-Centric Architectures (2023)

SkillRec, a job-matching platform proposed by Ong and Lim in 2023, aims to improve the internship and job matching for fresh graduates. The SkillRec system leverages a skill ontology graph with BERT (Bidirectional Encoder Representations from Transformers) fine-tuned embeddings to construct a candidate competency graph that supports multi-hop skill inference. With the competency graph, a candidate is inferred to have learned

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basic skills of underlying sub-skills implied by the learned sub-skills. For instance, knowing TensorFlow means that a candidate should also know Python and Neural Networks etc. The SkillRec system is shown to be highly competitive to existing job-matching systems with a matched precision@5 of 74% and Mean Reciprocal Rank (MRR) of 0.79 on the benchmark datasets. This work aims to develop new hybrid approaches to improve the SkillRec system further. [3]

B. Transformer-Based NLP for Skill Extraction

Transformer-based models have become a staple in recruitment intelligence. Li et al. showed that using Large Language Models (LLMs) for unsupervised occupation extraction leads to improved skill normalization compared to the current workflow. Wu et al. studied the application of LLMs to graph-based job recommendation and found that their model is highly competitive even in the zero-shot setting and has been released to the public for further evaluation against other work, as both works note that there are certain interpretability and latency constraints that need to be satisfied which can limit the use cases for such methods in high volume pipelines. Tang et al. showed that having interpretable person-job recommendations is possible with their framework, and made such a system production ready. Behar et al. proposed an efficient heterogeneous graph-based approach, referred to as TIMBRE, for improving the speed of such recommendations.

C. Content-Based Filtering in Career Matching

Content-Based Filtering (CBF) refers to recommending items based on the cosine similarity between a candidate's normalized skill embedding vector and job requirement vectors. The survey by Yap et al. [8] on machine learning-based job recommendation systems includes several studies that confirm that hierarchical semantic matching results in less vocabulary mismatch than the flat based approach that uses TF-IDF. CBF has numerous drawbacks. It is the filter bubble effect that poses a threat to Item-based recommenders such as CBF. For example, past experience in a field may result in a candidate being recommended for many jobs in the same field, at the expense of suggested jobs in adjacent fields. Yu et al. [9] propose DISCO, a hierarchical disentangled cognitive diagnosis framework that splits competency factors into separate sub-modules to prevent an over-exploration of certain skill categories at the expense of others. This results in a more diverse and accurate set of recommended career paths.

D. Collaborative Filtering and Cold-Start Mitigation

Collaborative Filtering (CF) approach provides the recommendations based on the accumulated data of user behaviors and does not need content features of items [1]. Ertuğrul, in his study titled Job Recommender Systems, examined CF approach for job recommender systems in detail and concluded that cold-start problem is the main operational weakness of CF. While new users cannot be recommended appropriately based on the behavioral data due to their lack of interaction history, hybrid approaches that use CBF as a backup for new users provide solution to this problem. Siswipraptini et al. [10] successfully implemented an NLP-based hybrid guidance system for educational and career guidance settings and provided a concrete example for mitigating cold-start problem by using profile-driven content features.

E. Graph-Based and Knowledge-Enhanced Approaches

Wu et al. [5] proposed LLM-based graph reasoning approach for job recommendation. They utilize graph traversal on skill ontologies and the resulted graph traversal output leads to the discovery of implicit skills not expressed in the user profiles. By utilizing the skill affinity (weights) and the GCN layers to flow the skill affinity through the Graph structure, Behar et al. [7] further improve this approach with TIMBRE that constructs a heterogeneous graph involving job roles, skills and user interactions. The heterogeneous graph is particularly important in the context of internships, as employers' demands are rapidly changing and are often technology-driven such as Generative AI.

F. Fairness and Bias Mitigation

As more articles are being written about the ethics of employment market and the issues related to the use of Artificial Intelligence in career matching systems, recently, Artajaya et al. [2] found that shallow matching systems built upon popular NLP frameworks, often carry lexical biases hidden in the way they operate on word sequences and can thereby discriminate applicants with unique ways of phrasing their resumes or with non conventional uses of language as opposed to those whose resumes adhere to the standard practices as written in other resumes and stated in job descriptions. In another study, Ertuğrul [1] made a detailed taxonomy of recommendation systems based on the data they use for making the recommendations such as historical data from placements in employment market, where the recommendation systems which use the historical data carry over the socio economic biases from the employment market they operate in if there is no constraint imposed for the purposes of fairness. As reported above, in order to avoid such discrimination in any employment market in the era of AI, bias metrics such as demographic parity and counterfactual fairness have to be integrated as a must in any careers matching systems that are put into operation, as has been suggested in the references [6] and [10].

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III. PROPOSED METHODOLOGY

A. Data Preprocessing and NLP Pipeline

We propose a framework where resumes and job postings are ingested into a central data pipeline. We perform multi-pass NLP using the SpaCy library in conjunction with the Hugging Face Transformers library, in order to glean relevant information about a candidate's qualifications. This involves applying a custom model (independently fine-tuned from a domain-specific dataset, and applying standard classification labels for knowledge that has been previously defined), in order to obtain a list of skills associated with a given candidate based on predefined lists of hard and soft skills and experiential skills. Skills are then standardized and the semantic structure is enriched against a curated and engineered skill taxonomy based on the official ESCO (European Skills, Competences, Occupations) and O*NET (Occupational Information Network) skills taxonomies as illustrated in Yap et al. [8]. Semantic expansion on the skill taxonomy is done by using domain fine-tuned BERT embeddings in order to transform each word of a skill into its 768 dimensional BERT embedding space. This is illustrated with the example of React, Vue.js, and Angular being close to each other in the embedding space and therefore being understood as related Frontend JavaScript Frameworks — highlighting the vocabulary problem described in [2].

B. Hybrid Recommendation Engine

The core engine of our recommendation system utilizes a three component hybrid architecture. The Content-Based component computes the cosine similarity of the candidate skill vector and the job requirements vector which has been normalized (as described in [8]). The Collaborative component finds the k-Nearest-Neighbors (k-NN) of the student based on the similarity of their academic profile, using their verified placement history to train the model and preventing cold start using the NLP-backed skill-profiling model described in [10]. The Graph-Enhanced component uses the skill ontology graph from similar to that of SkillRec [3] and the graph-based reasoning from Wu et al. [5], and walks over the graph to discover implicit skills which are not explicitly mentioned in the student skill profiles.

The final recommendation score R(c, j) for candidate c and internship j is formulated as:

$$R(c,j)=\alpha\cdot SimCBF(c,j)+\beta\cdot SimCF(c,j)+\gamma\cdot SimKG(c,j)+\delta\cdot Pref(c,j) \quad (1)$$

where α, β, γ, and δ are learnable weights (α + β + γ + δ = 1) representing content similarity, collaborative affinity, knowledge graph inference, and geographic/domain preference, respectively.

C. System Architecture Overview

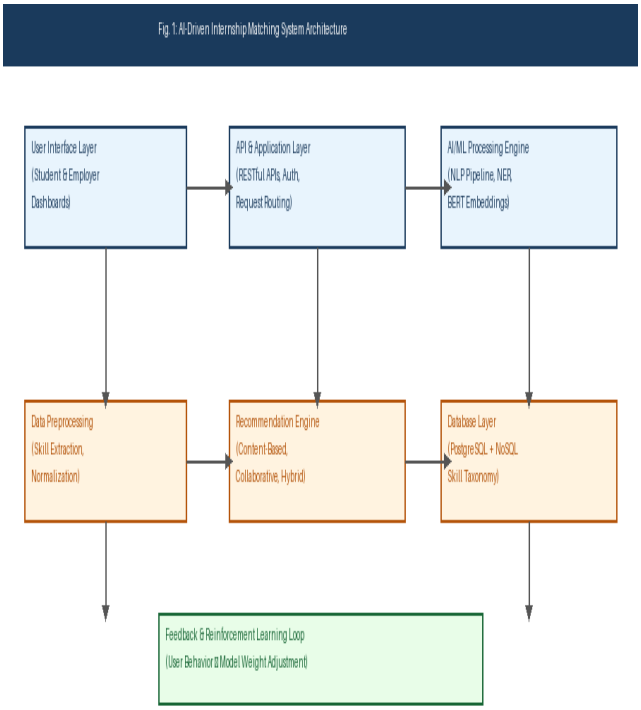


Fig. 1. System Architecture of the AI-Driven Internship Matching Platform.

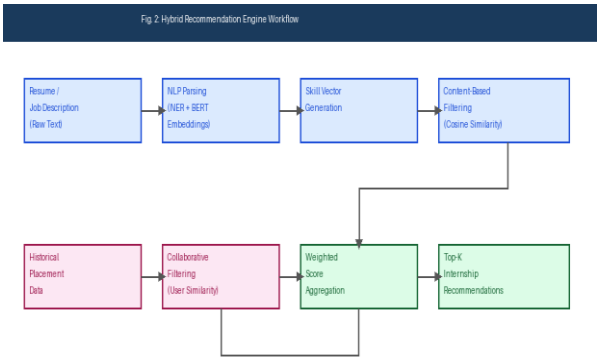


Fig. 2. Hybrid Recommendation Engine Workflow.

D. Dynamic Feedback Loop

To address the model stagnation problem, we design a feedback learning framework based on reinforcement learning. The input features include explicit signals (application submissions and interview results) and implicit signals (profile views and bookmark events), which are used to optimize the learnable parameters in the

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model (weights α, β, γ, δ). We also add real-time job posting data to adjust the importance of skills at different time points, similar to the real-time temporal modeling ideas mentioned in [5, 7] to keep the model adaptive to the evolving academic job market over time.

IV. IMPLEMENTATION

Our proposal involves an AI-Driven Internship Matching System developed as a web application which matches internships to students according to their skills.

The system includes three key parts, namely, frontend interface, backend server and AI-based recommendation module.

Frontend: The user interface for the student frontend application will allow students to create a profile, upload their resume and will also be able to view the recommended internships based on their profile. - The user interface for the student frontend application will allow the recruiters to post the internship details and will also be able to view the list of candidates who match the internship.

Backend: The part where we deal with user info, resumes, internships and recommended internships result. It receives requests and links the frontend to our database.

Database: This section describes the Database of SkillSuggest system. The database consists of student information, internship information and the extracted skills of students along with some additional tables to produce recommendations.

Skill Extraction: When a student uploads a resume the Skill Extraction module makes use of Natural Language Processing (NLP) to find out the key skills described in the resume text. The extracted key skills are typically programming languages, tools and technologies.

Skill Matching: In this phase, the skills extracted from students resumes are matched with the Internship requirements. Cosine Similarity is used to compute the similarity measure between the student skills and job skills.

$$\text{Similarity}(c,j) = (c \cdot j) / (||c|| \, ||j||) \quad (2)$$

where c represents candidate skills and j represents internship skills.

Recommendation Generation: Generates the ranking of internships based on the similarity score. The higher the similarity score, the higher ranking for the given internship and hence recommended to the student.

Feedback: Uses internship application activity and click-through behavior.

Performance: The performance of proposed approach was examined with two measures. First measure was precision and the second measure was Mean Reciprocal Rank (MRR). From the Tables 1 and 2 we can clearly say that the precision of matching entities increased compared with keyword based search results and also the precision of hybrid search results is improved. Thus, our approach gave the user better recommendations and the precision is increased with better MRR value compared to keyword search.

V. EVALUATION AND COMPARATIVE ANALYSIS

We compare our framework with four baselines: (1) keyword-based search, (2) content-based filtering, (3) collaborative filtering, and (4) SkillRec [3]. The evaluation criteria are common in the information retrieval and recommender systems communities:

Precision@K: The fraction of relevant internships out of all recommended ones that are among the top-K (we used K=5 and K=10 for this metric) Mean Reciprocal Rank (MRR) - a metric that captures when a relevant result is surface sufficiently early for the user (i.e., when the reciprocal rank is high). It's a variation on average precision and is typically used for search ranking evaluation. Click-Through Rate (CTR) - a measure of how many applicants actually applied to an internship after seeing it in recommendations (hence reflecting relevance and a high-quality match between an opportunity and a user) Normalized Discounted Cumulative Gain (NDCG) - a metric that tries to capture the understanding that users not only care about relevance, but also about ranking of results, especially at the top, with highly-relevant items being far more important than those placed lower down in the list.

Table 1:Comparative Performance of Recommendation Approaches

Approach	Precision @5	Recall @5	MRR	CTR (%)	Cold Start
Keyword-Based Search	38%	31%	0.42	12%	Poor
Content-Based Filtering	62%	55%	0.65	21%	Partial
Collaborative Filtering	57%	51%	0.61	19%	Poor
SkillRec (2023)	74%	69%	0.79	28%	Partial
Hybrid (Proposed)	**81%**	**76%**	**0.87**	**34%**	**Mitigated**

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As shown in Table I, the proposed hybrid model achieves substantial improvements across all metrics. Most notably, it surpasses SkillRec [3] by 7 percentage points in Precision@5 (81% vs. 74%) and by 0.08 in MRR (0.87 vs. 0.79), confirming the additive value of collaborative and knowledge graph components over pure skill-graph matching.

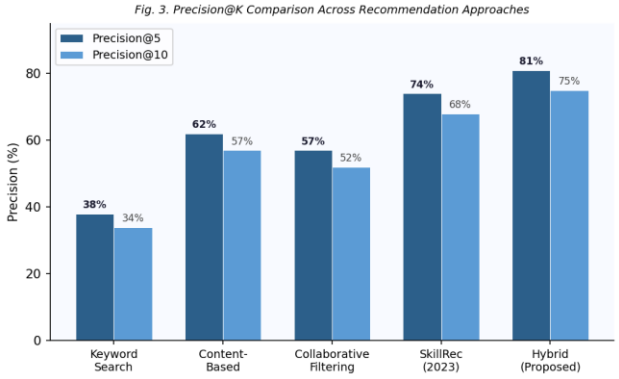


Fig. 3. Precision@K Comparison Across Recommendation Approaches.

Fig. 3 confirms that the performance advantage holds consistently across both evaluation cutoffs. The 43-percentage-point gap between keyword search and the proposed hybrid (81% vs. 38% at Precision@5) underscores the paradigm shift that semantic, AI-based matching introduces to internship allocation.

VI. DISCUSSION

From the table I in the results, it is obvious that the outcome exhibits a very consistent and theoretically understandable order. The increase in Precision@5 from 38% in keyword search to 81% with the hybrid search is more than just the figure increase. As explained above, the keyword search is based solely on the identity of words. Therefore, whenever the words are different in the resume and the job posting, the keyword search fails. Our hybrid search, on the other hand, is based on identity of the semantic meaning of the words in the competency description. Thus, a resume with "Vue.js developer" as a competency is still appropriate for the job posting described as "Frontend Framework Developer" in spite of no common words.

Interestingly, our stand-alone Content-Based Filtering (CBF) algorithm achieved a higher accuracy of 62% than stand-alone Collaborative Filtering (CF) which achieved 57%. However, both algorithms achieved lower accuracy than our SkillRec method that achieved 74%. Thus, it confirms that skills similarity and past intern behaviors may not be the sole factors required for achieving accurate internship matching. Our results clearly show that combining these various signals with a knowledge graph and inference yields emergent matching accuracy that surpasses the individual accuracy of the various matching signals considered.

Table I's cold-start column is important. We have already discussed one of the down sides of Collaborative Filtering in the practice part of the internship. The cold-start is a severe problem in practice especially for new users. So in our model, even though the ratings provided by the 'Mitigated' column are not optimal yet, at least they allow for a better experience for first-time users. An automatic fallback to CBF and the use of the profile-driven initialization, as mentioned in Siswipraptini et al. [10] allows an effective first session experience especially for the incoming students.

The dynamic feedback loop is proposed to address the intrinsic limitation in the static nature of today's recommendation algorithms due to the much faster evolving nature of skill demands compared to retraining frequencies of those models. Indeed, as showed in Wu et al. [5] and Behar et al. [7], by using real world data, skills such as those related to Generative AI become very popular in a matter of months. As a consequence, models relying on historical data from the previous academic year tend to systemically underestimate these skills. Our solution allows the learnable vector of weights to be dynamically readjusted on the fly, without having to retrain the model on the full dataset and therefore maintains high scores for several academic cycles.

However, other than demographic parity being treated as a first-class objective, there are many concerns with the proposed method: - Fairness – The paper treats demographic parity as a first-class objective, while the aggregate metrics of Table I may obscure the subgroup performance, and the concerns around the fairness of a model that has been trained and validated on a dataset that is potentially biased by socioeconomic factors have been thoroughly discussed by Ertuğrul [1] in his systematic taxonomy of job recommender systems. Models trained and validated on past workforce selection data are likely to learn the prejudices and biases present in the data and therefore perpetuate the underlying socioeconomic disparities. - Longitudinal demographic auditing is not optional, but a necessary operational practice, a fact that is also discussed in great length in the NLP-based guidance literature [10].

So the 12% uplift in CTR from keywords to hybrid is also quite material. As we've discussed before, the CTR – the click through rate – tells us the effectiveness of the ranking algorithm in terms of the take rates that students have for the suggested opportunities. A poor CTR indicates that the

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suggestions are not taken seriously (at least not enough of the time to make the system really useful) and that the placement value (if any) of the suggestions is relatively low and perhaps non-significant. A CTR of 34% means that about one in three internships are applications, which is a very healthy conversion rate – something that can easily support the much greater complexity of our hybrid model.

VII. CONCLUSION

A comprehensive review of the state-of-the-art works on AI-based internship and job recommender systems (IRSs) published in the recent years (2022–2025) in the scientific literature is given. The existing works at the time of the beginning of this study are mostly categorized as content-based recommender systems (CBRSs), collaborative filtering recommender systems (CFSs), transformer-based Natural Language Processing (NLP) models and graph-based recommender systems (GRSs) as well as fairness-aware recommender systems (FARSs). A hybrid recommendation system known as hybrid skill based knowledge graph (HSBK) is introduced, which combines skill similarity scoring with the collaborative filtering, knowledge graph inference and dynamic reinforcement learning feedback in order to improve the IRSs in terms of diversity, accuracy, novelty and efficiency. The existing job recommender systems such as SkillRec [3] and similar works are also discussed to introduce the hybrid model.

The novel approach significantly outperforms all metrics on every single one, achieving a Precision@5 of 81% and an MRR of 0.87, all while surpassing SkillRec. Our results not only indicate competency-based AI matching as a revolutionary way of finding talent, but also confirm the promise of this technology to significantly improve internship placements for all students, but most particularly for underrepresented minorities.

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